Mohammad Nazeri

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PhD Researcher in Embodied AI & Robotics, focused on bringing real-time planning into Generative World Models through multimodal representation learning that couples the planner directly to the model's own latent space. Proven track record of building advanced robot navigation systems, with hands-on experience deploying generative models onto physical robots.

EDUCATION

PhD in **Computer Science**, George Mason University **Jan. 2023 - Jan. 2027 (Exp.)**
MS in **Computer Science**, Institute for Advanced Studies in Basic Sciences (IASBS) **Sep. 2018 - Sep. 2021**
BS in **Computer Science**, Shahid Bahonar University **Sep. 2011 - Sep. 2015**

EXPERIENCE

Graduate Research Assistant (PhD) | RobotiXX Lab, George Mason University **Jan. 2023 - Present**

Research Focus: From Self-Supervised Perception to Generative World Models for Navigation. Spearheaded a progressive research agenda to build robust navigation systems:

- **VANP (Perception):** Pioneered self-supervised vision-action pre-training to enhance end-to-end perception modules for robotic navigation (IROS 2024).
- **VertiCoder (Dynamics):** Developed a self-supervised pre-training approach to learn complex robot kinodynamics (ICRA 2025).
- **VertiFormer (Unified System):** Combined perception and dynamics into a unified multi-task Transformer architecture utilizing multimodal inputs into a shared latent space to learn a complete navigation and dynamic prediction system (Under Review, RA-L).
- **Hydra (World Action Model):** Built a discrete-latent World Action Model that embeds planning directly in the model's latent space, achieving a 500× planning speedup over prior world models and 8-10/10 success rates across obstacle-rich physical trials on two robot platforms (Clearpath Jackal, Boston Dynamics Spot) (Under Review, ICLR).
- **Model Optimization:** Investigated knowledge distillation techniques to compress Large World Models into lightweight architectures for real-time, on-robot deployment.

Research Intern, Perception Team | Stack AV **May 2025 - August 2025**

- Enhanced the LaneDetectionTransformer for 3D lane detection by integrating video-based temporal cues, ensuring robust frame-to-frame consistency (e.g., maintaining lane classification despite visual fading) while meeting real-time inference constraints.
- Synthesized findings into a technical whitepaper to guide the team's long-term perception roadmap.

Research Intern, Augmented Intelligence Lab, McGill University **Feb. 2021 - Apr. 2022**

- Conducted a comprehensive meta-analysis of Unsupervised Learning in medical imaging, synthesizing trends from over 1,000 papers to identify key research gaps in label efficiency.
- PyTorch Instructor: MacMedHacks Summer School 2022

MS Student Researcher | Intelligent Systems Group, IASBS **Aug. 2018 - Aug. 2021**

- Developed a vision-based Mixture-of-Experts for autonomous driving systems in the CARLA simulator using imitation learning.
- Authored "Exploring reflective limitation of behavior cloning in autonomous vehicles," published at IEEE ICDM 2021 (20% acceptance rate).

Mentorship Experience

- **Alexandyr Card,** MS student, George Mason University (Co-author on VertiFormer and Hydra papers)
  - *Tentative Thesis Title:* Navigation World Models
- **Fatemeh Mehrara**, BS student, IASBS
  - *Bachelor's Thesis:* Smart City Challenges: Points of Interest Recommendation

Academic Service & Contributions

- **Peer review:**
  - Journals: IEEE Robotics and Automation Letters (RA-L), ACM/IEEE Transactions on Human-Robot Interaction (THRI).
  - Conferences: IEEE ICRA, IEEE/RSJ IROS, Conference on Robot Learning (CoRL), Robotics: Science and Systems (RSS).
- **Open Source Contributor:** scipy-lectures (Updated the fundamental NumPy tutorial).

AWARDS & HONORS

- **Best Student Presentation Award** | IEEE ICRA 2026, Vienna, Austria
  - Awarded at the 1st Workshop on Long-term Deployments in the Wild (LoWi) for the presentation of *VertiFormer*.
- **Microsoft Azure Compute Grant ($10,000)** | Microsoft Research Accelerator Program
  - Awarded $10,000 in cloud computing resources for project *Hydra*.

SKILLS

Core Competencies: Multimodal Deep Learning, Generative AI (Diffusion, Flow Matching, GANs), Vector Quantization & Discrete Representation Learning, Representation Learning, Self-Supervised Learning, Embodied AI
Languages: Python, C++, Lua, LaTeX
Frameworks & Libraries: PyTorch, ROS/ROS2, OpenCV, Hugging Face, Scikit-learn
Tools & Platforms: Linux, Docker, Git, CometML, W&B, Carla Simulator
Publications: 14+ papers in conferences (ICRA, IROS, ICDM). See Google Scholar for the full list.

SELECTED PUBLICATIONS (h-index: 11, i10-index: 11)

See Google Scholar for the full list of 14+ papers.

1. **Nazeri, M.,** Card, A., Huber, S., Pokhrel, A., Wang, Y., Hammele, R., Song, D., Pirk, S., & Xiao, X. (2026). Hydra: A Navigation World Action Model with Discrete Latent Planning and Continuous Flow-Matching Execution. Under review at ICLR 2027.
2. Pokhrel, A., Datar, A., **Nazeri, M.,** Cancelliere, F., & Xiao, X. (2026). Zero-Shot Adaptation to Robot Structural Damage via Natural Language-Informed Kinodynamics Modeling. Under review at ICRA.
3. Song, D., Le, N., Chen, J., Nazeri, M., Payandeh, A., Chandra, R., Mirsky, R., Mead, R., Xiao, L. and Xiao, X., 2026. HumAIN: Human-Aware Implicit Social Robot Navigation. 2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS).
4. **Nazeri, M.**, Pokhrel, A., Card, A., Datar, A., Warnell, G., & Xiao, X. (2025). VertiFormer: A Data-Efficient Multi-Task Transformer for Off-Road Robot Mobility. Under review.
5. Payandeh, A., Song, D., **Nazeri, M.**, Liang, J., Mukherjee, P., Raj, A.H., Kong, Y., Manocha, D. and Xiao, X., 2025, October. Social-LLaVA: Enhancing social robot navigation through human-language reasoning. In 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS).
6. **Nazeri, M.**, Datar, A., Pokhrel, A., Pan, C., Warnell, G., & Xiao, X. (2025). Verticoder: Self-supervised kinodynamic representation learning on vertically challenging terrain. 2025 IEEE International Conference on Robotics and Automation (ICRA).
7. **Nazeri, M.**, Wang, J., Payandeh, A., & Xiao, X. (2024). VANP: Learning Where to See for Navigation with Self-Supervised Vision-Action Pre-Training. 2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS).
8. **Nazeri, M.**, & Bohlouli, M. (2021). Exploring reflective limitation of behavior cloning in autonomous vehicles. 2021 IEEE International Conference on Data Mining (ICDM).